

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.1 TURNING EFFECT OF FORCE: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Think back to the Jembe Handle Puzzle from Lesson 1. In Kericho, Amina used a long-handled jembe (about 90 cm) and finished digging quickly and easily. Juma used a jembe with the same blade weight but a short handle (about 30 cm), and he struggled and tired quickly. Your task is to explain WHY, using the evidence you collected in this unit. Write your answers in the space under each prompt. Use full sentences, show every step of any calculation with units, and use the words pivot, force, perpendicular distance, moment, clockwise and anticlockwise correctly. Use $g = 10 \text{ N/kg}$ where you need weight. Write in your own words, as if explaining to a farmer in your home area.

Part 1: Finding the pattern in the metre-rule data

In the metre-rule investigation, a wooden metre rule was balanced on a pencil pivot at the 50 cm mark. Slotted masses were hung on each side. Here is the data from one group. Every trial was balanced. Use $W = mg$ with $g = 10 \text{ N/kg}$.

Trial	Left mass (g)	Left distance from pivot (m)	Right mass (g)	Right distance from pivot (m)
1	500	0.10	200	0.25
2	100	0.40	200	0.20
3	200	0.15	100	0.30

(a) Convert each mass to a weight (force) in Newtons. (b) Calculate $F \times d$ for the left side and for the right side in each trial, with units. (c) What pattern do you find? State it as a general rule in words and give it its scientific name. (d) Why is the metre rule balanced even though the masses and the distances on the two sides are different?

Write your answer here:

Part 2: Explaining the Jembe Handle Puzzle

Amina (handle about 90 cm) and Juma (handle about 30 cm) used jembes with identical blade weights to dig the same soil. (a) In the jembe, which point acts as the pivot, and what is the distance d ? (b) Assuming each of them applies the same effort force, use the equation $M = F \times d$ to explain why Amina found the digging easier than Juma. (c) By roughly how many times is Amina's handle length greater than Juma's? Use this ratio to compare the moments they produce. (d) A classmate says: 'Amina must have pushed with a much bigger force.' Use the idea of moments to respond to this claim.

Write your answer here:

Part 3: Calculating moments and the net turning effect

Look at this version of the jembe problem. Treat the jembe as able to turn about a fixed pivot P. A farmer grips the jembe 0.8 m from P and pushes with a force of 50 N, which turns the jembe anticlockwise about P. Friction at the blade applies a force of 20 N at a distance of 0.1 m from P, on the other side of P, which turns the jembe clockwise. (a) Calculate the anticlockwise moment and the clockwise moment, showing the equation and units. (b) Calculate the net moment. (c) Will the jembe rotate? If so, in which direction? (d) Why must you give each moment a direction (clockwise or anticlockwise) before you compare them? (e) When you work out a moment from a diagram, which distance must you use for d , and what common mistake do learners make when choosing it?

Write your answer here:

Part 4: Two hands on a steering wheel – couples

A matatu driver uses TWO hands on the steering wheel. In the suspended metre-rule demonstration, two equal forces were applied on opposite sides of the centre, pulling in opposite directions (anti-parallel), and the rule rotated. (a) What is a couple? (b) What is the net force on the rule from the two pulls, and what happens to the rule? Explain how this can be. (c) How do you calculate the moment of a couple? (d) Explain how a driver placing her hands further apart on the steering wheel helps her turn it, even though she does not push harder. (e) Give TWO other Kenyan examples of a couple.

Write your answer here:

Part 5: Answering the driving question

Driving question: Why does where you push, pull, or grip something determine how easy or hard it is to make it turn, and how do engineers and farmers in Kenya use this to design better tools?

(a) Write a full answer of 4–6 sentences. Use the jembe in Kericho as your main example and include the terms pivot, perpendicular distance, force and moment. (b) Give TWO other examples from our school walk-around or video clips (such as a gate, tap, crane at Mombasa port, pliers at Kamukunji, or a boda-boda brake in Kisumu). For each, say where the pivot is and how the tool's design makes turning easier. (c) Compare your first prediction from Lesson 1 with your explanation now. What has changed in your thinking?

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of $M = F \times d$ and the Principle of Moments	Calculates every moment correctly with units, converts masses to weights using $g = 10 \text{ N/kg}$, identifies clockwise and anticlockwise moments, and states the Principle of Moments accurately.	Calculates most moments correctly with units and states the Principle of Moments with minor gaps, such as a missing direction or unit.	Makes frequent calculation errors, omits units or directions, or cannot state the Principle of Moments.
Explaining the Jembe Handle Puzzle	Identifies the blade-soil contact as the pivot and explains that the longer handle (90 cm against 30 cm) gives a larger perpendicular distance and so a larger moment for the same force, rejecting the idea that a bigger force is needed.	Explains that a longer handle makes digging easier and refers to distance and moment, but the pivot or the same-force reasoning is incomplete.	Says only that a longer handle is easier or gives vague reasons, without linking to moment, pivot or perpendicular distance.

Understanding of couples and net moment	Defines a couple correctly, explains zero net force with pure rotation, uses $F \times d$ between the forces, and explains a real couple such as the two-handed steering wheel grip. Uses the perpendicular distance, not the arrow length.	Defines a couple and gives $F \times d$, with small gaps in explaining net force or the steering wheel example.	Confuses a couple with a single force, or does not explain why the object turns.
Connecting to Kenyan contexts and the driving question	Gives a coherent answer to the driving question with at least two correct Kenyan examples, each with pivot and design reason, and clearly shows how thinking changed since Lesson 1.	Gives a mostly complete answer with two examples, where the pivot or design reasoning is partly missing.	Gives one example or none, with little link to the driving question or to the change in thinking.

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